

Bringing Real Body as Self-Avatar into Mixed Reality: A Gamified Volcano Experience

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ABSTRACT

In this work we showcase a Mixed Reality experience able to bring users' own body as a self-avatar. This is mainly based on an algorithm that segment in real time the egocentric full body placing a regular stereo camera in front of a commercial Virtual Reality device. The work behind this paper is divided into: *i*) Data Acquisition; *ii*) Algorithm training and *iii*) Integration into a Mixed Reality environment. To the best of our knowledge, this is the first work that integrates real body as self-avatar both meeting real-time and segmentation quality in real-life conditions beyond benchmarking. We conclude the paper with some preliminary subjective evaluation measuring Presence and Embodiment Factors, that suggests the potential of using your own body as self-avatar in Mixed Reality.

1 INTRODUCTION

In the context of Virtual Reality (VR), different self-users representations (also known as self-avatars) have been studied due to their implication with respect to Sense of Presence [10] (SoP), and Sense of Embodiment (SoE) [2], interaction and cognitive load, etc.

One important aspect of self-user representations is their level of realism. Previous studies have already reported a positive correlation between virtual hand representation and the sense of ownership experienced by the user [1]. There has been also some works who attempted to incorporate real user's limb into the experience, by filtering egocentric video using color information that enable to easily incorporate hands, or even arms by using depth information.

In this paper, we propose to go beyond avatars or person representation based on upper limb bodies, and present a system able to incorporate the user's whole body, by segmenting RGB stereo video using a computer vision algorithm in real time. To the best of our knowledge, this is the first work that integrates real body as self-avatar both meeting real-time and segmentation quality in real-life conditions, along with a preliminary Quality of Experience feedback measuring Presence and Embodiment factors.

2 STATE OF THE ART

Recently the Mixed Reality community has started to explore the possibility of integrating users' own body into immersive experiences. Pigny *et al.* [9] integrated real users using a segmentation network with two decoders: one for segmenting egocentric users, the other for exocentric users. However their, segmentation execution speed could not currently totally keep up with the cameras update rate while maintaining a decent quality.

Later, in [6], the authors developed a system able to segment the user's hand and arms using a dataset composed of 43K images from TeGO, EDSH, and a custom EgoCam dataset, and a semantic segmentation deep learning network based on DeepLabv3+. For the

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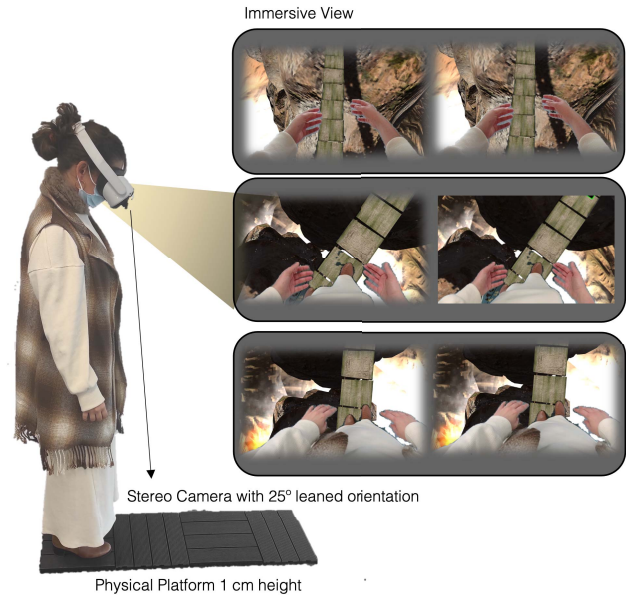


Figure 1: The user wears an Oculus Quest 2 and walks through a step, to experience in VR how she can see her own full body, walking in a bridge placed in the crater of a Volcano.

experiment, they used a HTC Vive HMD, a monocular RGB camera, and a Leap Motion controller to provide free-hands interaction camera. They capture 360x360 RGB video and manage to run Unity rendering and CNN segmentation in the same workstation.

3 EGOCENTRIC SEGMENTATION AND DATASET

The egocentric segmentation algorithm considered in this work is derived from a previous work [3]. It is based on Thundernet, a very shallow semantic segmentation algorithm composed of: *i*) an encoding subnetwork; *ii*) a pyramid pooling module that enables to extract features at different scales; and *iii*) a decoding subnetwork.

Equally important, is the dataset with which train the network and estimate their parameters. Attempting to have the most representative one, a combination of different egocentric datasets was used using: a subset from the semi-synthetic dataset [4], a subset of images from THU-READ dataset recently pixel-wise labelled [5], and an additional proprietary dataset, making a total of more than 8000 image pairs.

4 PROTOTYPE SYSTEM DESIGN AND IMPLEMENTATION

The entire experience includes a client workstation and a segmentation server. The client workstation includes an Intel Core i7-11800H and a Nvidia Geforce RTX 3070 GPU. A custom volcano virtual scene was created using Unity version 2019.3. The system was based on an Oculus Quest 2 head-mounted display (HMD) and a stereo camera which was hooked to the VR device using a custom

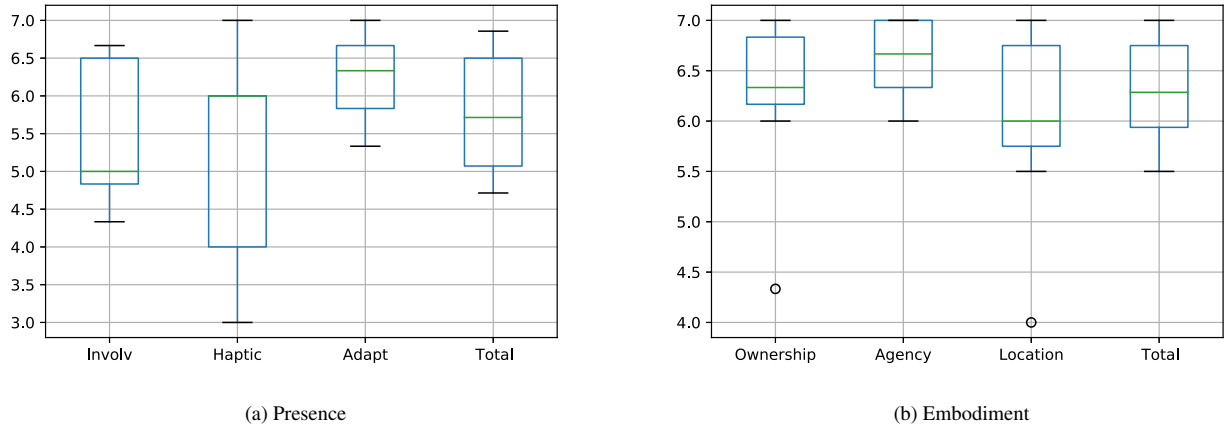


Figure 2: Results of SoP and SoE with boxplot, which represents within the box all data that lies between the 1st and 3rd quartile.

3D-printed attachment. The stereo frames were captured at a resolution of 1280x480 in Unity and streamed to the server using a TCP-based communication architecture, written as a C++ plugin interfacing Unity. The egocentric segmentation server was running on a PC with an Intel Xeon ES-2620 V4 @ 2.1GHz with 32 GB of RAM and powered with 2 GPU GTX-1080 Ti with 12GB RAM each. Several iterations and optimizations were done to reach a mean E2E latency (from capturing to display) of 45 ms, of which 22ms referred to the segmentation algorithm inference time.

5 IMMERSIVE EXPERIENCE AND PRELIMINARY RESULTS

Our study aims to investigate the role of full body video-based avatars in Mixed Reality. To this aim, a volcano immersive environment was created. Users were told to walk through a narrow bridge that crosses the crater of the volcano, forcing this way to pay attention to their lower limbs. To provide bridge passive haptics, we incorporate 1cm height steps, as can be seen in Fig.1. At the same time, and for interaction purposes, users had to collect a handful of virtual gold coins along the bridge using hand tracking capabilities provided by Oculus Quest 2.

The experience was tried by 8 subjects (5 men and 3 women, ages 21 – 54). Afterwards, users were asked to fill a short Embodiment [11] and Presence [8] questionnaires, based on Gonzalez-Franco and Peck EQ [7] and Witmer and Singer PQv3 [12]. Results, displayed on Fig. 2, show that our solution provides high values of avatar embodiment and sense of presence, even compared with a similar experience where only the arms could be perceived [11]. This suggests that perceiving your whole body within the immersive environment may have a strong positive impact in the overall realism of the experience.

6 CONCLUSIONS

In this work we have developed and subjectively tested the alternative of using user's own body as self-avatar, replacing computer generated avatars. This was achieved by the use of a stereo camera in front of commercial VR goggles and a real-time semantic segmentation algorithm. Preliminary results suggest the potential of using your own body for interacting in Mixed Reality. More in-depth analysis and comparison are needed to validate this statement.

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